

Approximating Power Series Centrality With The Katz Metric: Matching The Vertex Gap In Julia

Fotis Sofoulis¹[0009–0007–0619–5231], Konstantinos
Theodoropoulos²[0009–0001–1233–3019], Georgios
Drakopoulos²[0000–0002–0975–1877], Christos Troussas²[0000–0002–9604–2015], and
Phivos Mylonas²[0000–0002–6916–3129]

¹ CEID, University of Patras, Hellas
up1089341@upatras.gr

² ICE Department, University of West Attica, Hellas
ktheodorop@gmail.com, {georgedrakopoulos, ctrouss, mylonasf}@uniwa.gr

Abstract. Vertex centrality metrics are crucial in evaluating the role a single vertex or even a group of vertices play in structural terms as, for instance, in the the overall graph resilience. Such metrics being of higher order they reveal significant information. However, many of these metrics such as the total communicability are defined as a power series of the underlying graph adjacency matrix making their computation difficult from a computational, algorithmic, and numerical perspective. Therefore, it makes perfect sense to explore whether they can be approximated by centrality metrics which are less challenging. One such metric is Katz centrality which in its core relies on solving a special linear system involving said adjacency matrix. In order to assess the approximation quality, statistical measures including Kendall’s tau and Spearman’s rho are computed for two power series centrality metrics, namely the exponential and the Mercator series, for a number of benchmark graphs including ca-CondMat and ca-AstroPh and for a number of Katz parameters determined by linear algebraic criteria. The results indicate that the proposed approximation is feasible.

Keywords: Vertex centrality · Katz centrality · Resolvent centrality · Higher order metrics · Matrix exponential · Graph mining · Julia.

1 Introduction

Vertex centrality metrics rank how important vertices are in terms of a given structural metric, or in some cases a combination of structural and functional attributes especially in the context of social media. Common structural attributes include the number of triangles a given vertex participates in and the number of shortest paths crossing a certain vertex, whereas functional features include the flow crossing a given vertex. An important class of vertex centrality metrics is the one consisting of a power series of the underlying graph adjacency matrix which frequently provide considerable insight at the expense of elevated computational

cost, increased algorithmic complexity, or potential numerical challenges. For instance, a prime example of algorithmic complexity is that the matrix exponential can be computed in a number of ways including the matrix series of equation (15) and the matrix Laplace transform of equation (16).

The above issues can be addressed in a number of ways. The most common computational way is to use parallelism like Julia³, which can be done through coroutines, multi-threading, distributed computing, or GPU computing, or concurrency like the goroutines⁴ in Go, which are lightweight functions managed by the Go runtime system. Another strategy, which is the principal motivation for this work, is algorithmic in nature and lies in the direction of replacing the power series with another centrality metric as long as the approximation is acceptable under a pre-specified criterion. When a solution is obtained in the form of vertex ranking, then it can be directly used as is or driven to a more complex system, for instance an iterative solver such as conjugate gradient, as initial input.

Table 1. Notation synopsis.

Symbol	Meaning	First in
$\triangleq$	Definition or equality by definition	Eq. (1)
$\mathbf{I}_n$	$n \times n$ identity matrix	Eq. (3)
$\mathbf{e}_k$	k -th column of the identity matrix	Eq. (14)
$\mathbf{1}_n$	$n \times 1$ vector of ones	Eq. (21)
$\text{diag}[d_1, \dots, d_n]$	Diagonal matrix with elements $d_1, \dots, d_n$	Eq. (3)
$\ \cdot\ $	Matrix or vector norm	Eq. (25)
$\text{tr}(\cdot)$	Matrix trace	Eq. (26)
$\tilde{\mathbf{x}}$	Transform of vector $\mathbf{x}$	Eq. (6)
$\mathcal{L}[\cdot]$	Laplace transform	Eq. (16)

Refining the above, the primary research objective of this conference paper is the exploration of the fitness of approximating the total communicativity and the Mercator series, two vertex centrality metrics matrix based on the exponential and the logarithm power series respectively, with the Katz centrality, which relies on the resolvent of a shifted version of the adjacency matrix. To assess the approximation fitness three statistical measures as well as the relative error are used over four common values of the shift parameter, each applied to six graphs. The implementation language was Julia⁵ which is designed for computationally intensive engineering scenarios. In the context of this work the term *vertex gap*, a reference to the absurdist term *mineshaft gap*⁶ from the 1964 film *Dr. Strangelove*⁷, denotes the divergence in ranking between any two metrics.

³ <https://docs.julialang.org/en/v1/manual/parallel-computing>

⁴ <https://go.dev/tour/concurrency/1>

⁵ <https://julialang.org>

⁶ <https://www.rand.org/pubs/commentary/2023/05/beyond-the-mine-shaft-gap-navigating-threat-perceptions.html>

⁷ <https://www.imdb.com/title/tt0057012/>

The remainder of this work is structured as follows. In section 2 the relevant scientific literature regarding vertex centrality, digital influence, and graph mining is briefly overviewed. The proposed implementation of the vertex centrality metrics is described in section 3 and the results obtained by them are discussed in 4. In section 5 the major findings of this work are recapitulated and possible future research directions are given. Bold capital letters denote matrices, small bold vectors, and small scalars. Technical acronyms are explained the first time they are encountered in the text. The terms *attribute* and *feature* are used interchangeably. Finally the notation of this work is summarized in table 1.

2 Related Work

Graph mining [4] is a vast field encompassing diverse problems like graph clustering [22], graph compression [23], and parallel graph generation [6]. Graph mining has a broad spectrum of applications such as cyber threat intelligence [15], scene generation [19], mining imbalanced data [21], discovering community structure in Twitter [8], mining rules for knowledge graphs through large language models (LLMs) [18], and factorizing the adjacency matrix [10]. Graphical ML has garnered considerable research interest through GNNs [24] whose applications include hyperspectral image classification [26], drug discovery [25], and social network affective analysis [16]. The connections of Mittag-Leffler functions to network theory are explored in [2]. Moreover, computations can be accelerated by offloading them to GPU [9] or by efficient parallel connectivity checks [5].

Vertex centrality metrics [1] are computed based on graph structural features such as vertex degrees [27], vertex percolation [20], and the number of shortest paths crossing a given vertex [17]. Additionally, there are edge centrality metrics [13]. Applications include vertex classification [3], recommendation systems based on multilayer graphs [7], player classification based on the Bartle taxonomy [12], and electrical grid anomaly detection [14]. Also, centrality can be used to cluster graphs constructed from attributes extracted by the MBTI [11].

3 Centrality Metrics

3.1 Matrix power series

In this subsection the general theory of matrix power series is briefly overviewed. A matrix power series is defined for a square matrix $\mathbf{A}$ in the general case as in equation (1). Essentially the latter represents a matrix function. Examples of such functions include matrix exponential, matrix sine, matrix square root, and matrix polynomials. Additionally, an important consequence of defining and building matrix functions is the Cayley-Hamilton theorem which for an invertible matrix casts the inverse as a polynomial of the original matrix.

$$g(\mathbf{A}) \triangleq \sum_{k=0}^{+\infty} \gamma_k \mathbf{A}^k = \gamma_0 \mathbf{I} + \gamma_1 \mathbf{A} + \gamma_2 \mathbf{A}^2 + \dots \quad (1)$$

Given the form of equation (1), it makes perfect sense to ask whether it converges. In the general case, its convergence depends on the following factors:

- The number of terms in the power series. A finite number of terms guarantees convergence and keeps the sum bounded and well-defined.
- The rate decay of coefficients $\{\gamma_k\}$. A sufficiently quick decay compared to the increase of elements of $\mathbf{A}^k$ will ensure the power series convergence.
- The spectral properties of $\mathbf{A}$. If the spectral radius of $\mathbf{A}$ is less than one, then progressively the eigenvalues of $\mathbf{A}^k$ shrink and the latter tends to zero.

An equivalent form of equation (1) can be computed assuming that $\mathbf{A}$ has a known spectral factorization and, moreover, that it has distinct eigenvalues which simplifies the structure of the eigenvalue matrix $\mathbf{\Lambda}$ and at the same time eliminates the need for generalized eigenvectors keeping eigenvector matrix $\mathbf{S}$ orthogonal. Then the spectral factorization of $\mathbf{A}$ takes the form of (2) below.

$$\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^T \in \mathbb{R}^n \quad (2)$$

The eigenvector matrix $\mathbf{S}$ of equation (2) is orthogonal, forming thus a base of $\mathbb{R}^n$ in which $\mathbf{A}$ is diagonal. Repeated eigenvalues would result in the eigenvalue matrix $\mathbf{\Lambda}$ having a Jordan block form instead of being diagonal.

$$\mathbf{S}^T\mathbf{S} = \mathbf{I}_n \quad \text{and} \quad \mathbf{\Lambda} \triangleq \text{diag}[\lambda_1, \dots, \lambda_n] \quad (3)$$

Under the same assumptions using the spectral factorization of (3) results in rewriting equation (1) in the alternative form of equation (4). By exploiting the orthogonality of $\mathbf{S}$ the power series structure commutes with the eigenvector matrix and it is applied on the eigenvalue matrix. Because of the structure of the latter, the power series structure is applied to each of the diagonal elements of $\mathbf{\Lambda}$. Thus only the middle factor of equation (2) needs to be changed in order to obtain $g(\mathbf{A})$, which is a major algorithmic and computational advantage.

$$g(\mathbf{A}) = \sum_{k=0}^{+\infty} \gamma_k \mathbf{S}\mathbf{\Lambda}^k \mathbf{S}^T = \mathbf{S} \text{diag}[g(\lambda_1), \dots, g(\lambda_n)] \mathbf{S}^T = \mathbf{S} g(\mathbf{\Lambda}) \mathbf{S}^T \quad (4)$$

A direct consequence of equation (4) is that the solution of an arbitrary linear system can be rewritten as in equation (5). By changing the basis and exploiting the diagonal form of $g(\mathbf{\Lambda})$ in the new basis the spectral solution follows.

$$g(\mathbf{A})\mathbf{x} = \mathbf{b} \Leftrightarrow \mathbf{S}g(\mathbf{\Lambda})\mathbf{S}^T\mathbf{x} = \mathbf{b} \Leftrightarrow \mathbf{x} = \mathbf{S}g(\mathbf{\Lambda})^{-1}\mathbf{S}^T\mathbf{b} \quad (5)$$

Observe that equation (5) can be equivalently written like (6). The latter means that it is a diagonal system in the space spanned by the eigenvectors of $\mathbf{A}$. If computing $\mathbf{S}$ is easy, which is frequently the case when $\mathbf{A}$ has a special structure, then it is preferable to obtain the spectral solution $\tilde{\mathbf{x}}$ first and then return to the standard basis in order to get the solution vector $\mathbf{x}$.

$$\mathbf{S}^T\mathbf{x} = g(\mathbf{\Lambda})^{-1}\mathbf{S}^T\mathbf{b} \Leftrightarrow \tilde{\mathbf{x}} = g(\mathbf{\Lambda})^{-1}\tilde{\mathbf{b}} \quad (6)$$

Matrix $g(\mathbf{\Lambda})$ is constructed by applying the function $g(\cdot)$ to the eigenvalues of $\mathbf{A}$ as shown in equation (7). This follows from equation (4) as the power series of equation (1) is commuted with the spectral expansion of $\mathbf{A}$.

$$\begin{aligned}
g(\mathbf{\Lambda}) &= \begin{bmatrix} \sum_{k=0}^{+\infty} \gamma_k \lambda_1^k & 0 & 0 \dots 0 & 0 \\ 0 & \sum_{k=0}^{+\infty} \gamma_k \lambda_2^k & 0 \dots 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 \dots 0 & \sum_{k=0}^{+\infty} \gamma_k \lambda_n^k \end{bmatrix} \\
&= \begin{bmatrix} g(\lambda_1) & 0 & 0 \dots 0 & 0 \\ 0 & g(\lambda_2) & 0 \dots 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 \dots 0 & g(\lambda_n) \end{bmatrix} \\
&= \text{diag}[g(\lambda_1), g(\lambda_2), \dots, g(\lambda_n)]
\end{aligned} \tag{7}$$

Because of its structure the diagonal matrix of equation (5) can be easily inverted as shown in equation (8). Please observe that diagonality is preserved.

$$g(\mathbf{\Lambda})^{-1} = \text{diag}\left[\frac{1}{g(\lambda_1)}, \frac{1}{g(\lambda_2)}, \dots, \frac{1}{g(\lambda_n)}\right] \tag{8}$$

If one of the eigenvalues of $\mathbf{A}$ is a root of $g(\cdot)$, then it follows that the system matrix of equation (6) is singular. In this case a regularized version $\mathbf{\Delta}$ as in equation (9) of $g(\mathbf{\Lambda})$, its pseudoinverse, or the Moore-Penrose can be used to obtain an approximate solution based on some optimality criterion.

$$\mathbf{\Delta} = \delta_0 \mathbf{I}_n + g(\mathbf{\Lambda}) \tag{9}$$

Matrix $\mathbf{\Delta}$ is the coefficient matrix corresponding to the following least squares (LS) problem of equation (10). The latter seeks a solution maintaining a balance between fitting the available data and smoothness. This balance is sought through the hyperparameter δ_0 which regulates the fluctuation of the solution as well as the relative weight of the second term of equation (10) with respect to the first one. Moreover, δ_0 determines the eigenvalue shift of $g(\mathbf{\Lambda})$.

$$\tilde{\mathbf{x}}_{LS} = \text{argmin} \left[\left\| g(\mathbf{\Lambda}) \tilde{\mathbf{x}} - \tilde{\mathbf{b}} \right\|_2 + \delta_0 \|\tilde{\mathbf{x}}\|_2 \right] \tag{10}$$

If μ is an eigenvalue of $g(\mathbf{\Lambda})$ with corresponding eigenvector $\mathbf{u}$, then the corresponding eigenvalue of $\mathbf{\Delta}$ can be computed as shown in equation (11). Because of the special nature of the eigenvector $\mathbf{u}$, the non-linear structure of the matrix on the left hand is applied only to the eigenvalue.

$$\mathbf{\Delta} \mathbf{u} = (\delta_0 \mathbf{I}_n + g(\mathbf{\Lambda})) \mathbf{u} = \delta_0 \mathbf{u} + \mu \mathbf{u} = (\delta_0 + \mu) \mathbf{u} \tag{11}$$

As stated earlier the spectral solution of (5) is advantageous to be computed when $\mathbf{S}$ can be easily found so that moving from the standard basis to the

transformed one and back is efficient. For instance, the eigenvector matrix of a Toeplitz matrix is the discrete Fourier transform matrix. Recall that a Toeplitz matrix has the structure shown in equation (12).

$$\mathbf{T} \triangleq \begin{bmatrix} t_{1,1} & t_{1,2} & \dots & t_{1,n} \\ t_{1,n} & t_{1,1} & \dots & t_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ t_{1,1} & t_{1,2} & \dots & t_{1,n} \end{bmatrix} \quad (12)$$

As another specific example consider the discrete Laplace matrix of equation (13) which appears among others in the solution of the Laplace equation over a discretized grid and has applications to graph theory through the related graph Laplacian matrix which is a measure of information diffusion in the graph.

$$\mathbf{M} \triangleq \begin{bmatrix} -2 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 & -2 \end{bmatrix} \quad (13)$$

In general, the Laplacian matrix is strongly associated with the discretized second derivative in continuous scenarios or the second order difference in discrete cases. Because of its importance, its linear algebraic properties like its spectrum and eigenvectors have been extensively studied.

3.2 Power series centrality

In this subsection some major power series centrality metrics are explored. According to this broad class of metrics, each vertex v_k is assigned as centrality score $\mathbf{g}[k]$ the k -th element of the inverse of a matrix function $f(\cdot)$ defined on the graph adjacency matrix as shown in equation (14).

$$\mathbf{g}[k] \triangleq \mathbf{e}_k^T (f(\mathbf{A}))^{-1} \mathbf{e}_k = (f(\mathbf{A}))^{-1}[k, k] \quad (14)$$

One of the most well studied matrix series is that of matrix exponential shown in equation (15). This exponential matrix plays a central role in signal processing and control theory among other fields. From a graph theoretic perspective, the matrix exponential contains the number of paths between every possible number of vertices where paths get an inverse factorial weight to heavily prioritize locality, assessing thus the overall graph resilience in purely structural terms. This is why it is called the *total communicability*.

$$e^{\mathbf{A}} \triangleq \sum_{k=0}^{+\infty} \frac{1}{k!} \mathbf{A}^k = \mathbf{I}_n + \mathbf{A} + \frac{1}{2} \mathbf{A}^2 + \dots \quad (15)$$

It should be incidentally highlighted that the Laplace transform of the exponential of matrix $\mathbf{A}$ is its resolvent as shown in equation (16). This equation is

fundamental in signal processing and control theory and recently has been also been of the mainstays of graph signal processing (GSP).

$$\mathcal{L}[e^{\mathbf{A}t}] = (s\mathbf{I}_n - \mathbf{A})^{-1} \quad (16)$$

Another matrix series used to assess vertex centrality is the Mercator series of equation (17). In general the Mercator expansion is used to project cylindrical surfaces on parallelograms while maintaining the distances under appropriate nonlinear scaling. Thus, local shapes remain accurate. As a centrality metric it is interesting that paths of odd length have negative contribution.

$$\ln(\mathbf{I}_n - \mathbf{A}) \triangleq \sum_{k=0}^{+\infty} \frac{(-1)^k}{k} \mathbf{A}^k = \mathbf{I}_n - \mathbf{A} + \frac{1}{2}\mathbf{A}^2 - \frac{1}{3}\mathbf{A}^3 + \dots \quad (17)$$

The Neumann series of equation (18) corresponds to repeated application of the adjacency matrix $\mathbf{A}$ on itself. Unless $\mathbf{A}$ is a contraction operator, or equivalently it has a spectral radius equal to or larger than one, this series diverges. In the context of graph paths, there is no weight and thus each path contributes equally to the global vertex centrality.

$$(\mathbf{I}_n - \mathbf{A})^{-1} = \sum_{k=0}^{+\infty} \mathbf{A}^k = \mathbf{I}_n + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots \quad (18)$$

The Neumann series is similar to the geometric series of equation (19) defined on the real scalar variable x . The geometric series converges only when the absolute value of x is strictly less than one as it can be seen by techniques such as the Raabe convergence criterion. Still, it should be noted that under other convergence definitions like the Cesaro criterion, the geometric series converges.

$$\frac{1}{1-x} = \sum_{k=0}^{+\infty} x^k = 1 + x + x^2 + x^3 + \dots, \quad |x| < 1 \quad (19)$$

As a sidenote, the Neumann power series appears modified in the multistep Richardson method of equation (20), an iterative system for solving linear systems through repeated shifted versions of the original coefficient matrix. Therein $\mathbf{A}$ and $\mathbf{b}$ denote the coefficient matrix and the right hand side respectively.

$$\mathbf{x}^{[k+1]} = \left(\mathbf{I}_n - \omega^{[k]}\mathbf{A}\right)\mathbf{x}^{[k]} + \omega^{[k]}\mathbf{b} \quad (20)$$

In general, repeated matrix applications are rooted in operator theory and in the context of graph theory they can be interpreted as successive hops in a given graph. As such, there are also connections to Markov chains.

3.3 Katz centrality

The Katz vertex centrality metric is defined as in equation (21). Therein the centrality score each vertex v_k receives is the k -th entry of the solution of the

linear system. Although computationally this is a different way to assign centrality scores to vertices, the same higher order reasoning applies on an abstract level: Each vertex receives a score based on that assigned to vertices close to it, where vertices located farther in the graph are penalized heavier through the *dampening factor* or *Katz parameter* α to prevent undue influence. Notice that the shifted adjacency matrix appears also in (17) and (18). The right hand side vector of all ones represents a limiting state where reaching every vertex from every other one. In other words, it is the scaled uniform probability vector when a random walker is allowed to do infinite steps in a connected graph.

$$(\mathbf{I}_n - \alpha \mathbf{A})^{-1} \mathbf{g} = \mathbf{1}_n \quad (21)$$

The optimal Katz parameter α is determined as in equation (22). It is derived by a minimax process on a recursive function of the adjacency matrix. The maximum eigenvalue of the adjacency can be computed or approximated, perhaps through a combination of the power method and the Rayleigh quotient.

$$\alpha_{\text{deg}} \triangleq \frac{1 - e^{-\lambda_{\max}}}{\lambda_{\max}} \quad (22)$$

Another option for the Katz parameter is that of equation (23), which is derived as an approximation to (23) under mild assumptions regarding the spectrum of $\mathbf{A}$ which hold for many important classes of adjacency matrices. Still, the maximum eigenvalue needs to be computed or estimated. The Katz parameter is inversely proportional to it, indicating that the more spread the eigenvalues are, then the less the influence of neighboring and nearby vertices should be.

$$\alpha_{0.85} \triangleq \frac{0.85}{\lambda_{\max}} \quad (23)$$

A third option for the Katz parameter is that of equation (24). This approximation comes from a different approach rooted in the recursive solution of large linear systems, especially in adaptive signal processing. In this case, said approximation places an upper bound on the learning parameter of adaptive algorithms such as steepest descent and least means squares (LMS) where the eigenvalues are computed over the input vector autocovariance matrix.

$$\alpha_{0.5} \triangleq \frac{1}{2 \lambda_{\max}} \quad (24)$$

The final option for the Katz parameter is based on the infinity norm of $\mathbf{A}$, which is the average degree scaled. Compared to the previous approaches, it can be more easily computed or estimated.

$$\alpha_{\text{deg}} \triangleq \frac{1}{1 + \|\mathbf{A}\|_{\infty}} \quad (25)$$

The relationship between the spectrum of matrix and its trace is established through equation (26). The infinity norm approximates the trace, especially

for sparse matrices like these of scale free graphs, and in turn the maximum eigenvalue from below. Scale free graphs appear in numerous field such as brain connectivity, scientometrics, social media, and transportation networks.

$$\text{tr}(\mathbf{A}) \triangleq \sum_{k=1}^n \mathbf{A}[k, k] = \sum_{k=1}^n \lambda_k \leq n \lambda_{\max} \quad (26)$$

The goodness of this approximation depends heavily on the concentration of the matrix spectrum and any properties influencing it. Therefore, large gaps in it lead to looser bounds, whereas repeated eigenvalues lead to tighter ones.

4 Results

In order to evaluate the titular vertex gap, namely how close the Katz centrality is to that computed by the exponential and the Mercator series. In order to do this, a number of statistical and linear algebraic criteria have been used over six benchmark graphs. The latter include the Zachary karate club, a very small graph with some interesting structural points, and Power grid, which is rich in structural patterns because of the peculiarities involved in energy distribution such as load balancing. CondMat and AstroPh come from scientometrics and reflect the conditions inherent in contemporary scientific collaboration.

The criteria used to evaluate the approximation similarity are the following:

- Kendall’s τ .
- Spearman ρ .
- Pearson’s r .
- The κ_1 condition number.
- The relative error e_r .

The above metrics as well as the vertex centrality schemes were implemented in Julia, which has a vast library of linear algebraic methods readily available.

The results for the six graphs and the exponential series are given in table 2, while those for the Mercator series are given in table 3. From the above tables it can be seen that in general the Mercator series is better approximated with the Katz centrality when the optimal Katz parameter is chosen. However, when spectral information about the graph adjacency matrix is not available, then the parameter based on the average degree provides a satisfactory approximation, even if it is the worse of the four examined here.

5 Conclusions

This conference paper focuses on the feasibility of approximating matrix power series vertex centrality metrics with the Katz centrality metric. Typically the former metrics rely heavily on a combination of local and global structural connectivity patterns such as vertex degree or the number of triangles a given vertex

Table 2. Results for the exponential series.

Graph	Katz	τ	ρ	r	κ_1	error
Zachary	α_{opt}	0.9946	0.9995	0.4792	5.46e3	0.9909
	$\alpha_{0.85}$	0.9855	0.9979	0.4789	3.70e1	0.9901
	$\alpha_{0.50}$	0.9901	0.9989	0.4794	7.050e	0.9881
	α_{deg}	0.9865	0.9985	0.4793	4.40e0	0.9874
Football	α_{opt}	0.9994	0.9997	0.8698	1.37e5	0.9992
	$\alpha_{0.85}$	0.9931	0.9969	0.8699	1.55e1	0.9998
	$\alpha_{0.50}$	0.9882	0.9995	0.8699	3.30e1	0.9993
	α_{deg}	0.9921	0.9996	0.8698	1.3301	0.9997
Les Miserables	α_{opt}	0.9978	0.9994	0.5374	1.25e6	0.9998
	$\alpha_{0.85}$	0.9912	0.9993	0.5376	4.30e1	0.9996
	$\alpha_{0.50}$	0.9933	0.9996	0.5375	8.60e0	0.9998
	α_{deg}	0.9922	0.9993	0.5376	4.48e0	0.9998
Power grid	α_{opt}	0.9931	0.9998	0.2506	1.22e4	0.9634
	$\alpha_{0.85}$	0.9927	0.9998	0.2507	3.69e1	0.9619
	$\alpha_{0.50}$	0.9911	0.9997	0.2506	6.40e0	0.9582
	α_{deg}	0.9908	0.9997	0.2506	4.33e0	0.9576
ca-CondMat	α_{opt}	0.9977	0.9999	0.0558	2.13e12	0.9998
	$\alpha_{0.85}$	0.9963	0.9993	0.0557	1.34e2	0.9997
	$\alpha_{0.50}$	0.9962	0.9998	0.0558	1.67e1	0.9994
	α_{opt}	0.9962	0.9994	0.0558	4.50e0	0.9992
ca-AstroPh	α_{opt}	0.9987	0.9992	0.1064	4.20e18	0.9992
	$\alpha_{0.85}$	0.9984	0.9999	0.0754	1.40e2	0.9998
	$\alpha_{0.50}$	0.9983	0.9999	0.0753	1.95e1	0.9997
	α_{opt}	0.9985	0.9997	0.0754	4.44e0	0.9999

Table 3. Results for the Mercator series.

Graph	Katz	τ	ρ	r	κ_1	error
Zachary	α_{opt}	0.9931	0.9956	0.4914	4.33e0	0.9917
	$\alpha_{0.85}$	0.9819	0.9852	0.4913	2.37e0	0.9921
	$\alpha_{0.50}$	0.9795	0.9806	0.4915	3.21e0	0.9955
	α_{deg}	0.9741	0.9801	0.4909	3.61e0	0.9996
Football	α_{opt}	0.9844	0.9749	0.5512	2.34e0	0.9947
	$\alpha_{0.85}$	0.9733	0.9682	0.5511	4.11e0	0.9931
	$\alpha_{0.50}$	0.9728	0.9605	0.5514	3.65e0	0.9917
	α_{deg}	0.9704	0.9585	0.5516	5.55e0	0.9895
Les Misérables	α_{opt}	0.9795	0.9599	0.5384	4.49e0	0.9933
	$\alpha_{0.85}$	0.9740	0.9488	0.5384	5.71e0	0.9959
	$\alpha_{0.50}$	0.9654	0.9463	0.5386	3.22e0	0.9989
	α_{deg}	0.9602	0.9421	0.5380	3.1200	0.9999
Power grid	α_{opt}	0.9619	0.9587	0.4321	3.83e0	0.9912
	$\alpha_{0.85}$	0.9587	0.9545	0.4324	3.12e0	0.9915
	$\alpha_{0.50}$	0.9543	0.9487	0.4325	2.93e0	0.9950
	α_{deg}	0.9510	0.9466	0.4326	2.18e0	0.9984
ca-CondMat	α_{opt}	0.9731	0.9533	0.4813	3.14e0	0.9814
	$\alpha_{0.85}$	0.9702	0.9484	0.4811	3.54e0	0.9922
	$\alpha_{0.50}$	0.9693	0.9451	0.4817	4.01e0	0.9945
	α_{deg}	0.9631	0.9315	0.4813	4.55e0	0.9968
ca-AstroPh	α_{opt}	0.9912	0.9845	0.5619	4.13e0	0.9817
	$\alpha_{0.85}$	0.9899	0.9819	0.5618	0.5618	0.9825
	$\alpha_{0.50}$	0.9856	0.9713	0.5621	2.99e0	0.9856
	α_{deg}	0.9822	0.9699	0.5614	3.99e0	0.9831

and their computation requires the computation of a matrix function as well as its inversion, rendering them thus expensive and potentially numerically susceptible to instability. The Katz centrality, customizable through a hyperparameter, is a viable alternative as it requires the solution of a single linear system. The approximation quality is assessed using statistical criteria like Kendall’s tau and Person’s r as well as linear algebraic metrics like the condition number. Results using a range of benchmark graphs including bibliographical ones and the Zachary karate club are encouraging. The implementation was in Julia, a language aiming at data intensive applications with an extensive library.

This conference paper can be extended in a number of ways. First and foremost, more benchmark graphs can be used. Moreover, other approximations can be used, for example a weighted or a rational approximation of simpler centrality metrics. Alternatively, truncated series or incomplete factorizations can be used.

Acknowledgements This work is part of Project 451, a long term research initiative with a primary objective of developing novel, scalable, numerically stable, and interpretable higher order analytics.

References

1. Aprahamian, M., Higham, D.J., Higham, N.J.: Matching exponential-based and resolvent-based centrality measures. *Journal of Complex Networks* **4**(2), 157–176 (2016)
2. Arrigo, F., Durastante, F.: Mittag–Leffler functions and their applications in network science. *SIAM Journal on Matrix Analysis and Applications* **42**(4), 1581–1601 (2021)
3. Bei, Y., Chen, W., Chen, H., Zhou, S., Yang, C., Fan, J., Huang, L., Bu, J.: Correlation-aware graph convolutional networks for multi-label node classification. In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*. pp. 37–48 (2025)
4. Bhattacharya, R., Rathore, Y.K., Tran, T.A., Swarnkar, S.K.: *Graph Mining*. Springer (2026)
5. Cao, W., Meng, K., Lin, Z., Tan, G.: GLumin: Fast connectivity check based on LUTs for efficient graph pattern mining. In: *Proceedings of the 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming*. pp. 455–468 (2025)
6. Dong, B., Wang, W., Liu, X., Wang, Y., Yin, D.: GraphCon: A parallel graph construction from relational data. *Big Data Mining and Analytics* **9**(2), 448–464 (2026)
7. Drakopoulos, G., Giannoukou, I., Sioutas, S., Mylonas, P.: Self organizing maps for cultural content delivery. *NCAA* **31**(7) (2022). <https://doi.org/10.1007/s00521-022-07376-1>
8. Drakopoulos, G., Giotopoulos, K.C., Giannoukou, I., Sioutas, S.: Unsupervised discovery of semantically aware communities with tensor Kruskal decomposition: A case study in Twitter. In: *SMAP. IEEE* (2020). <https://doi.org/10.1109/SMAP49528.2020.9248469>

9. Drakopoulos, G., Liapakis, X., Tzimas, G., Mylonas, P.: A graph resilience metric based on paths: Higher order analytics with GPU. In: ICTAI. pp. 884–891. IEEE (2018). <https://doi.org/10.1109/ICTAI.2018.00138>
10. Drakopoulos, G., Mylonas, P.: A genetic algorithm for Boolean semiring matrix factorization with applications to graph mining. In: Big Data. IEEE (2022). <https://doi.org/10.1109/BigData55660.2022.10020828>
11. Drakopoulos, G., Mylonas, P.: Clustering MBTI personalities with graph filters and self organizing maps over Pinecone. In: Big Data. IEEE (2024). <https://doi.org/10.1109/BIGDATA62323.2024.10825637>
12. Drakopoulos, G., Voutos, Y., Mylonas, P.: Annotation-assisted clustering of player profiles in cultural games: A case for tensor analytics in Julia. BDCC **4**(4) (2020). <https://doi.org/10.3390/bdcc4040039>
13. Durón, C., Kravitz, H., Brio, M.: Edge percolation centrality: A new measure to quantify the influence of edges during percolation in networks. PLOS one **20**(9) (2025)
14. Elnour, M., Atat, R., Takiddin, A., Ismail, M., Serpedin, E.: Eigenvector centrality-enhanced graph network for attack detection in power distribution systems. Electric Power Systems Research **241** (2025)
15. Freitas, S., Gharib, A.: Web scale graph mining for cyber threat intelligence. In: ACM SIGKDD Conference on Knowledge Discovery and Data Mining. vol. 2, pp. 4447–4456. ACM (2025)
16. Karavokyris, M., Sioutas, S.: Graph neural networks for affective social media: A comprehensive overview. In: CIKM Workshops. CEUR-WS (2022)
17. Kitamura, Y.K.N., Izumi, T.: Of approximate vertex ranking by betweenness centrality. In: Structural Information and Communication Complexity: 32nd International Colloquium, SIROCCO 2025, Delphi, Greece, June 2–4, 2025, Proceedings. p. 431. Springer Nature (2025)
18. Luo, L., Ju, J., Xiong, B., Li, Y.F., Haffari, G., Pan, S.: Chatrule: Mining logical rules with large language models for knowledge graph reasoning. In: Pacific-Asia Conference on Knowledge Discovery and Data Mining. pp. 314–325. Springer (2025)
19. Min, Y., Yang, M., Zhang, J., Wang, Y., Wu, A., Deng, C.: Vision-language interactive relation mining for open-vocabulary scene graph generation. In: International Conference on Computer Vision. pp. 16755–16764. IEEE/CVF (2025)
20. Nijhara, P., Venkatachalam, L., Singh, A.H., Chandramouli, A., Jana, S., Kothapalli, K., Banerjee, D.S.: Efficient parallel algorithms for dynamic percolation centrality. In: Proceedings of the 54th International Conference on Parallel Processing. pp. 309–319 (2025)
21. Qi, Y., Lu, Q., Dou, S., Sun, X., Li, M., Li, Y.: Graph neural network-driven hierarchical mining for complex imbalanced data. In: 8th International Symposium on Big Data and Applied Statistics (ISBDAS). pp. 320–324. IEEE (2025)
22. Ren, T., Zhang, H., Wang, Y., Ju, W., Liu, C., Meng, F., Yi, S., Luo, X.: MHGC: Multi-scale hard sample mining for contrastive deep graph clustering. Information Processing & Management **62**(4) (2025)
23. Schatz, K., Beasley, J.M., Tropsha, A., Chirkova, R.: GAME+: Lossless knowledge-graph compression for improving the efficiency and effectiveness of rule mining. International Journal of Data Science and Analytics **21**(1), 79 (2026)
24. Xu, L., Peng, J., Jiang, X., Chen, E., Luo, B.: Graph neural network based on graph kernel: A survey. Pattern Recognition **161** (2025)

25. Zhang, O., Lin, H., Zhang, X., Wang, X., Wu, Z., Ye, Q., Zhao, W., Wang, J., Ying, K., Kang, Y., et al.: Graph neural networks in modern AI-aided drug discovery. *Chemical Reviews* **125**(20), 10001–10103 (2025)
26. Zhao, X., Ma, J., Wang, L., Zhang, Z., Ding, Y., Xiao, X.: A review of hyperspectral image classification based on graph neural networks. *Artificial Intelligence Review* **58**(6) (2025)
27. Zhao, X., Fu, S., Yang, R., Yang, L., Chen, Y., Zhang, J., Long, J., Zhou, F., Zhao, Y.: Investigating visual perception of degree centrality in graph visualization. *IEEE Transactions on Visualization and Computer Graphics* (2025)